

## # ROS2 for ATOSFleetManagement Cheat Sheet

### ## 1) Source environment

```
```bash
source /opt/ros/humble/setup.bash
source ~/atos_ws/install/setup.bash
```
```

### ## 2) Build ATOSFleetManagement packages

```
```bash
cd ~/atos_ws
colcon build --packages-select atos atos_gui --symlink-install
source install/setup.bash
```
```

### ## 3) Start ATOSFleetManagement (normal)

```
```bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True
```
```

### ## 4) Start ATOSFleetManagement with 3 simulators

```
```bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True with_truck_simulator:=
True
```
```

#### ### Default simulator setup in launch

```
- `L5S-TRUCK-SIM-1`: `start_index=0`, `target_speed_kmh=80`, `ignore_warning_speed_c
ommands=True`
- `L5S-TRUCK-SIM-2`: `start_index=250`, `target_speed_kmh=40`
- `L5S-TRUCK-SIM-3`: `start_index=500`, `target_speed_kmh=40`
```

### ## 5) Open GUI

```
- URL: `http://localhost:8420`
- Go to tab: `RuralRoad Map`
- UI shows live trucks + `Distance To Next Truck Ahead`
```

### ## 6) Run in Docker (same image, with/without simulators)

```
```bash
cd ~/Documents/repos/ATOS
docker compose -f docker-compose-fleetmanagement.yml up --build
```
```

#### With simulators:

```
```bash
WITH_TRUCK_SIMULATOR=True docker compose -f docker-compose-fleetmanagement.yml up --
build
```
```

#### Without simulators:

```
```bash
WITH_TRUCK_SIMULATOR=False docker compose -f docker-compose-fleetmanagement.yml up -
-build
```
```

#### Start without rebuild (if image already built):

```
```bash
```

```
docker compose -f docker-compose-fleetmanagement.yml up
```
```

Start in background:

```
```bash
docker compose -f docker-compose-fleetmanagement.yml up -d
```
```

When to use `--build`:

- Use `--build` only when image content changed (Dockerfile, dependencies, or source copied into image).
- For normal restart, do **\*\*not\*\*** use `--build`.

## 7) Run Docker as a systemd service

```
```bash
sudo mkdir -p /opt/atos
sudo rsync -a ~/Documents/repos/ATOS/ /opt/atos/
sudo cp /opt/atos/scripts/atosfleetmanagement.env.example /etc/default/atosfleetmanagement
sudo cp /opt/atos/scripts/atosfleetmanagement.service /etc/systemd/system/atosfleetmanagement.service
sudo systemctl daemon-reload
sudo systemctl enable --now atosfleetmanagement
```
```

Service controls:

```
```bash
sudo systemctl status atosfleetmanagement
sudo systemctl restart atosfleetmanagement
sudo journalctl -u atosfleetmanagement -f
```
```

## 8) Copy to another server (what files are needed)

Recommended (safest):

```
```bash
rsync -a ~/Documents/repos/ATOS/ user@<server>:/opt/atos/
```
```

Minimum required for Docker-based ATOSFleetManagement:

- `Dockerfile`
- `docker-compose-fleetmanagement.yml`
- `scripts/run\_atosfleetmanagement.sh`
- `scripts/installation/`
- `atos/`
- `atos\_gui/`
- `atos\_interfaces/`
- `conf/`
- Optional for systemd service:
  - `scripts/atosfleetmanagement.service`
  - `scripts/atosfleetmanagement.env.example`

## 9) Check running nodes

```
```bash
ros2 node list
```
```

## 10) Check key topics

```
```bash
```

```
ros2 topic list | rg "truck_objects|speed_command"
```
```

## 11) Watch live truck states for GUI

```
```bash
ros2 topic echo /atos/truck_objects/state
```
```

## 12) Watch control speed commands

```
```bash
ros2 topic echo /atos/truck_objects/speed_command
```
```

## 13) TruckObjectControl COT interfaces

- ROS topic input: `/atos/truck\_objects/cot` (placeholder format)
- TCP input: `0.0.0.0:8114` listener in TruckObjectControl
- Truck clients connect to: `127.0.0.1:8114` (same machine) or `:8114`
- CoT `` is in `m/s` (GUI displays `km/h`)

## 14) Placeholder ROS COT test (manual)

```
```bash
ros2 topic pub /atos/truck_objects/cot std_msgs/msg/String "data: 'id=truck_1;distance_m=1000;tcp_connected=1;lat=57.78357;lon=12.76389;speed_mps=2.8;course_deg=150'" -
1
```
```

## 15) Start one simulator manually (custom test)

```
```bash
ros2 run atos atos_truck_simulator --ros-args \
  -p uid:=L5S-TRUCK-UBUNTU \
  -p start_index:=300 \
  -p target_speed_kmh:=35.0 \
  -p acceleration_mps2:=0.6 \
  -p tcp_host:=127.0.0.1 \
  -p tcp_port:=8114
```
```

## 16) Simulator option: ignore first-limit slowdown

Use this only on selected trucks:

```
```bash
-p ignore_warning_speed_commands:=true
```
```

Behavior:

- Ignores non-zero warning command (for example `30 km/h` at `< 400 m`)
- Still obeys stop command (`0 km/h` at `< 200 m`)

## 17) Fleet control thresholds

- Gap `< 400 m` => publish `target\_speed\_kmh=30`
- Gap `< 200 m` => publish `target\_speed\_kmh=0`

## 18) Stop everything

- Press `Ctrl+C` in launch terminal